A STRICTLY SINGULAR NONCOMPACT COMPOSITION OPERATOR

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AI assistance was used in adapting and checking the argument and the exposition.

Laitila, Nieminen, Saksman, and Tylli ask in Problem 4.3(4): "Is there a Banach space E of scalar-valued analytic functions on $\mathbb{D}$ and an analytic map $\varphi : \mathbb{D} \to \mathbb{D}$, for which $C_\varphi \in \mathcal{S}(E) \setminus \mathcal{K}(E)$?" Here $\mathcal{S}(E)$ denotes the strictly singular operators and $\mathcal{K}(E)$ the compact operators.

Laitila, Nieminen, Saksman, and Tylli, *Rigidity of composition operators on the Hardy space H^p*, arXiv:1607.00113.

Answered: yes. Evidence: Sketch.

Yes. One can take $\varphi(z) = z^2$.

Proof. Write every positive integer uniquely as $2^k j$ with j odd, and represent $f(z) = a_0 + \sum_{j \text{ odd}} a_{0,j} z^j + \sum_{k\geq 1, j \text{ odd}} a_{k,j} z^{2^k j}$. Define

$$\|f\|_E = |a_0| + \sum_{j \text{ odd}} |a_{0,j}| + \left(\sum_{k\geq 1, j \text{ odd}} \frac{|a_{k,j}|^2}{(k!)^2 j^{2(k-1)}} \right)^{1/2}.$$

This is a Banach direct sum of a scalar, an ℓ^1 space, and a weighted ℓ^2 space. If $n = 2^k j$ with $k \geq 1$, then $|a_{k,j}| \leq \|f\|_E k! j^{k-1} \leq \|f\|_E n^{2k} \leq \|f\|_E \exp(2(\log n)^2/\log 2)$. Hence the power series converges locally uniformly in $\mathbb{D}$, so E is a Banach space of analytic functions; finite coefficient sequences, and therefore polynomials, are dense.

We use that the formal inclusion $J : \ell^1 \to \ell^2$ is strictly singular but not compact. The unit vectors show it is not compact. To see strict singularity, let $M \subset \ell^1$ be infinite-dimensional. For any N and small $\delta > 0$, choose unit vectors $u_1, \dots, u_N \in M$ with successive disjoint coordinate intervals carrying all but δ of each vector's ℓ^1 mass. Their interval restrictions y_i have disjoint supports and $\|u_i - y_i\|_1 < \delta$. For $v = N^{-1} \sum_i u_i$, this gives $\|v\|_1 \geq 1 - 2\delta$ and $\|Jv\|_2 \leq N^{-1/2} + \delta$. Letting N grow and then δ decrease shows that J is not bounded below on any infinite-dimensional subspace.

For $\varphi(z) = z^2$, composition fixes the scalar coordinate, maps the $k = 0$ odd layer by J into the $k = 1$ Hilbert layer, and on normalized coordinates $b_{k,j} = a_{k,j}/(k! j^{k-1})$ for $k \geq 1$ acts by

$$b_{k,j} \longmapsto \frac{b_{k,j}}{(k+1)j}.$$

The latter is compact because its diagonal weights tend to zero; the scalar action has finite rank. The remaining layer map is strictly singular: if it were bounded below on an infinite-dimensional subspace of E, its first-layer projection would be infinite-dimensional and J would be bounded below there. Finally, adding the compact pieces preserves strict singularity: approximate the compact part by a finite-rank map, restrict to the kernel of that map, and use strict singularity of the layer map to contradict any proposed lower bound.

The odd monomials z^j have norm one, and their images z^{2j} are separated unit vectors in the $k = 1$ Hilbert layer. Thus C_φ is not compact. This is the requested example. □

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